
SurvCalc
User Manual v1.2-2011-09-28

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1. INTRODUCTION

1.1 Overview

SurvCalc is a C++ computer program which analyses data from stratified random surveys. Its primary purpose is to calculate estimates of biomass and/or length frequencies (LFs), and associated coefficients of variation (c.v.s), from survey data. These data may be held either in a database structured like the Ministry of Fisheries database *trawl* (Mackay 2000) or in flat files. SurvCalc supersedes, and uses some code from, the program 'trawlsurvey' (Vignaux 1994).

Users of SurvCalc are urged to include their input files in an appendix to any report describing the analysis of stratified random surveys. The main input file for SurvCalc has been designed so that, taken together with this manual, it fully documents all the choices the user makes in calculating biomass etc (e.g., the choice of stations to include, and how distance towed is calculated if there is no recorded value). This will allow readers of survey reports to replicate the analyses therein. When SurvCalc is run using data from flat files, rather than from a database, these flat files should also be included in the report to complete the documentation.

Each time SurvCalc is run it carries out one of the seven following tasks. The first three tasks involve different types of calculations that may be made either during a survey or afterwards. Each can be applied to analyse multiple species in multiple surveys (or trips) in a single run of SurvCalc, and the species analysed may be different in different trips.

1. Task `calc_biomass`. Calculates biomasses, by stratum and overall. Can also calculate biomasses for sub-populations defined by sex and/or length range (e.g., for males of length between 20 cm and 80 cm). C.v.s are calculated for all biomasses. Optionally, calculates, during a survey, projected biomass c.v.s (i.e., the c.v.s expected at the end of the survey given the data to date – this can be useful during a 2-phase survey).

2. Task `calc_LFs`. Calculates LFs by station and/or stratum and/or overall. All LFs are presented by sex (including a category for unsexed) and overall. The user can choose between five alternative methods of scaling the LFs. C.v.s are not calculated for LFs.

3. Task `calc_biomass_and_LFs`. Combination of tasks `calc_biomass` and `calc_LFs` but only one method of LF scaling is allowed (scaling to represent estimated numbers in the population) and c.v.s are optionally calculated for LFs by stratum and sex.

The next task will usually be used at sea at (or near) the end of phase 1 of a 2-phase survey (Francis 1984). It can be applied only to a single trip (but can involve multiple species) and is intended to provide information useful in deciding on the phase-2 allocation (i.e., how a specified number of phase-2 stations should be allocated amongst the survey strata). Some guidance on how SurvCalc should be used during a 2-phase survey is given in Section 6.

4. Task `phase_2_calc`. Calculates, separately for each species requested, the relative gains (in terms of reduced variance of biomass estimates) associated with allocating varying numbers of phase-2 stations in each stratum. From this information the optimum phase-2 allocation can be derived for each species.

The last three tasks simply reorganise the survey data and output it in a different form.

5. Task `output_flat_files`. Output data in one or more of seven types of flat file. Depending on the type, each line of an output flat file may represent a stratum, a station, a

catch or subcatch record (i.e., a combination of a station and a species), or a length record (i.e., a combination of station, species, subcatch, and length).

6. Task `output_LW_coeffs`. Output a table of length-weight coefficients. This shows what length-weight coefficients are held in database `rdb` for each species so that the user can decide whether to use these stored coefficients or to specify new coefficients. (See Section 5.1.6 for a description of how these coefficients are used in various calculations).

7. Task `output_for_catch_at_age`. Output a file (in either ‘survey’ or ‘survey.sub’ format, whichever is appropriate) for input to the catch-at-age software (Bull & Dunn 2002). That is a file that can be read by the catch-at-age function `import.length.data`.

The remainder of this section compares `SurvCalc` with its predecessor, `trawlsurvey`, and discusses some possible future extensions of `SurvCalc`. Section 2 describes how to run `SurvCalc`; Sections 3 and 4 describe the various input and output files, respectively; Section 5 documents the calculations in `SurvCalc`; and Section 6 discusses how `SurvCalc` output should be used in 2-phase surveys.

1.2 Relationship of `SurvCalc` to `trawlsurvey`

This section is aimed at past users of the program `trawlsurvey` (Vignaux 1994) and may safely be ignored by others. It is intended to help introduce these past users to the main features of `SurvCalc` by comparing it with the earlier program.

From the user’s point of view a major difference between `trawlsurvey` and `SurvCalc` is the way in which they define the analyses they want. For `trawlsurvey` this was done by entering information via a series of blue screens, whereas for `SurvCalc` this information is written into the main input file (in a command-block format similar to that for `CASAL`). This file, together with the `SurvCalc` manual, will serve as a complete documentation of the analysis.

A second important difference concerns the computers on which each program will run and their data requirements. `trawlsurvey` runs only on Unix machines and requires the survey data to be in an *Empress* database structured like *trawl* (Mackay 2000). In contrast, `SurvCalc` runs on both Unix and Windows machines and can access survey data either in flat files or in *Empress* or *Postgresql* databases.

A third, more subtle, difference between the two programs is that the format of the output has been tweaked so that it is easier to read it into R for plotting and further analysis (see Appendix 2).

All the main calculations in `trawlsurvey` – of biomass and LFs – are exactly the same in `SurvCalc` (in fact much `trawlsurvey` code was reused in `SurvCalc`). However, `SurvCalc` includes several new features (see Section 1.2.1), discards a few features of `trawlsurvey` (see Section 1.2.2), and corrects a couple of minor errors in that program (see Section 1.2.3).

Before describing these differences in functionality of the two programs it’s worth noting, for the record, some technical programming differences. `trawlsurvey` is actually the combination of two programs: an *Empress* 4GL interface, which is what the user sees (this generates all the blue screens), and a C program, which is run, in batch mode, from that interface. `SurvCalc` is a single C++ program.

1.2.1 Extensions to trawlsurvey

The following are the main features of SurvCalc that were not possible with trawlsurvey (excluding those just mentioned).

- Biomass and LFs can be calculated for multiple trips and/or species in a single run.
- Data can be extracted from the *trawl* database and output in flat files of station data (one line per station), stratum data (one line per stratum), catch data, and length data.
- Input files for use in the catch-at-age software can be output
- The analysis of potting surveys is more straightforward and sensible (i.e., the user will not have to make up fake values for doorspread and distance towed).
- Sex-specific length-weight coefficients are allowed.
- The calculations for phase 2 of a 2-phase survey are much more extensive (see Section 6).
- The user can control the degree of precision (expressed as the number of significant figures and/or decimal places) of each type of output.

1.2.2 Excluded features of trawlsurvey

1. trawlsurvey produces LFs as percentages (in the main output file) and as numbers (in separate files, but some summary information for the numbers LFs is, confusingly, included in the main output file). SurvCalc produces LFs only as numbers (LFs as percentages are easily calculated from these).

2. SurvCalc does not allow the user to define bounds and interval for LFs (e.g., for lengths 20 cm to 50 cm in 2 cm steps). All SurvCalc LFs cover the full range of the data in 1-cm steps. The length bounds are not well handled in trawlsurvey: it is not made clear that (oddly) they apply to the percentage LFs and (confusingly) the summaries of the numbers LFs, but not to the numbers LFs; also, the user is not informed when there are length data outside the specified length bounds.

3. trawlsurvey outputs a table containing, *inter alia*, mean fish densities and biomass estimates by stratum, where the density units – kg/km and kg/km² – are chosen by the user, although the biomass estimates are always based on densities in kg/km². This is potentially misleading because the obvious inference from this table is that the biomass estimates derive from the presented densities. In SurvCalc, the densities in this table are always in kg/km² (but kg/km can be calculated, if requested, and output to separate station-catch and/or stratum-catch files).

1.2.3 Corrections to trawlsurvey

1. LFs for stations and species with more than one subcatch are not well handled in trawlsurvey. For such stations there can be more than one record in the *trawl* table *t_lgth* with the same *station_no*, *species*, and *lgth*, as in the following extract.

trip_code	station_no	species	subcatch_no	lgth	percent_samp	no_a
tan0601	79	HOK	1	56	5.53	2
tan0601	79	HOK	2	56	10.55	8

In trawlsurvey this will, misleadingly, produce two lines for length 56 cm in the LF for station 79 (but there's no problem with stratum and overall LFs). This does not occur with SurvCalc.

2. Although the trawlsurvey blue screen says that there should be no overlap in the length ranges of sub-populations, the program actually does allow overlap, and sometimes this causes errors (e.g., when, accidentally, two identical length ranges were specified, trawlsurvey produced just one sub-population biomass for this length range but this biomass was too high by a factor of 2). SurvCalc allows overlap in length ranges and treats these correctly.

1.3 Possible future extensions to SurvCalc

This section describes features that may be incorporated in future versions of SurvCalc, depending on demand from users (and the coding effort required).

1. SurvCalc could be extended to analyse survey data in databases like *scallop* and *oyster*. The structures of these databases are broadly similar to *trawl*, but with some relatively minor differences that would have to be allowed for.
2. When designing new trawl surveys, there is a need to decide, on the basis of previous survey data, how many stations need to be allocated to each stratum to achieve a target c.v.. One way this is currently done is in the following two steps: (a) extract data from previous surveys in the *trawl* database, and (b) run the Splus function *allocate* (see Appendix 2). It might be useful to combine these two steps in SurvCalc.
3. SurvCalc allows the calculation of biomasses of sub-populations defined by sex and length (e.g., all males less than 30 cm long). This could be extended to allow the use of gonad-stage data in defining sub-populations (e.g., all females of stage > 2). This would involve using *trawl* tables *t_fish_bio* and/or *t_lgth_stage*, not currently used by SurvCalc.
4. SurvCalc could calculate length-weight coefficients for a species in a survey (or surveys). Ideally, this calculation should (a) be robust to outliers, (b) include graphical output to show the user how well the data fit the estimated curve and what range of lengths is well covered by the relationship, and (c) include the ability to test for significant differences between the parameters for males and females.
5. Some users have asked for the ability to calculate total biomass for large groups of species (e.g., 'all fish', which means excluding invertebrates etc). This might involve using the attribute *descrptn* in table *species_master* in database *rdb* to define groups of species.
6. Current options for LF scaling make no allowance for correlation in the samples (i.e., the fact that, typically, the lengths of two fish from the same tow are more similar than those from different tows). When more sophisticated scaling schemes are developed they should be available in SurvCalc.

2. RUNNING SURVCALC

SurvCalc is run from the command prompt (in Windows or Unix) by typing a command like `SurvCalc -b -t stnfile > myfile`. It uses information from the main input file and possibly some flat files (all described in Section 3), makes certain calculations (documented in Section 5), and writes results to the main output file (`myfile` in the preceding example), and possibly other files (see Section 4). In the command line, between `SurvCalc` and `>`, there must be one or more *run-time arguments* (which may occur in any order) as described below.

The command line must include *exactly one* of following run-time arguments, which describes the task required of SurvCalc:

Argument	Task	Task description
-b	calc_biomass	calculate biomass
-l	calc_LFs	calculate LFs
-B	calc_biomass_and LFs	calculate biomass and LFs
-2	phase_2_calc	do calculations for allocating phase-2 stations in a 2-phase survey
-o	output_flat_files	only output flat files of data (to be used only in conjunction with arguments -s, -t, -u, or -v)
-p	output_LW_coeffs	output a table of length-weight coefficients (from database <code>rdb</code>)
-c	output_for_catch_at_age	output a file (in either 'survey' or 'survey.sub' format, whichever is appropriate) for input to the catch-at-age software
-h	help	list all arguments
-L	show license	display the SurvCalc end user license

In addition, one or more of the following run-time arguments can be used to provide information about input and output files:

Argument	Description
-S [infile]	read stratum data from a flat file rather than a database; default value for infile is <code>stratum.in</code>
-T [infile]	read station data from a flat file rather than a database; default value for infile is <code>station.in</code>
-U [infile]	read catch data from a flat file rather than a database; default value for infile is <code>catch.in</code>
-V [infile]	read length data from a flat file rather than a database; default value for infile is <code>lgth.in</code>
-W [infile]	read subcatch data from a flat file rather than a database; default value for infile is <code>subcatch.in</code>
-X [infile]	read combined station and catch rate data from a flat file rather than a database; default value for infile is <code>station_catch.in</code>
-s [outfile]	output a flat file of stratum data (one line per stratum); default value for outfile is <code>stratum.out</code>
-t [outfile]	output a flat file of station data (one line per station); default value for outfile is <code>station.out</code>
-u [outfile]	output a flat file of catch data (one line per catch record); default value for outfile is <code>catch.out</code>
-v [outfile]	output a flat file of length data (one line per length record); default value for outfile is <code>lgth.out</code>

-w [outfile]	output a flat file of subcatch data (one line per subcatch record); default value for outfile is <code>subcatch.out</code> ; to be used only with -o
-x [outfile]	output a flat file of combined station, catch, and catch rate data (one line per station); default value for outfile is <code>station_catch.out</code> ; to be used only with -b, -B, or -l
-y [outfile]	output a flat file of combined stratum, catch, and catch rate data (one line per stratum); default value for outfile is <code>stratum_catch.out</code> ; to be used only with -b, -B, or -l
-f infile	alternative name for the main input file (e.g., -f <code>ORH.slc</code> means that the main input file will be <code>ORH.slc</code> rather than the default <code>input.slc</code>)

SurvCalc obtains the data to be analysed either from an (Empress or PostgreSQL) database, or from flat files (-S, -T,-U,-V, -W, and -X), but not both (e.g., you cannot provide station and stratum flat files but expect SurvCalc to get catch and length data from the database). The flat files that need to be provided depend on the run time task and some of the preferences specified in the input file (e.g., biomass calculation involving no sub-populations and using the ‘recorded’ catch preference requires stratum (-S), station (-T), and catch data files (-U), or alternatively stratum (-S) and station-catch (-X) data files). SurvCalc will give error messages if data provided are inconsistent with the run time task and preferences.

3. INPUT FILE SPECIFICATIONS

SurvCalc requires a main input file (described in Section 3.1) which describes the data that are to be analysed and specifies some details of the analyses and the desired output. The actual data to be analysed are read either from an (Empress or PostgreSQL) database or from user-provided flat files (described in Section 3.2).

3.1 The main input file

The main input file for SurvCalc has default name `input.slc` (but this name can be changed if run-time argument -f is used – see Section 2). It uses a command-block format similar to that used in CASAL. The order of command blocks within the main input file, and of subcommands within a command block, is arbitrary (other conventions of this format are described in Appendix 1).

The commands used in the main input file fall into five categories, depending on their function, as follows.

Commands defining the data (Section 3.1.1). What data are to be analysed: what type of survey (trawl or pot); which trip and species are to be analysed; which stations to use from that trip; which database (if any) should be accessed and what additional data should be extracted.

Commands modifying the data (Section 3.1.2). What changes should be made to the data extracted from the database: stations can be reassigned to different strata (either existing ones, or new user-defined strata); areas of existing strata can be changed.

Commands extending the data (Section 3.1.3). Non-database information needed for the analyses: vulnerability and vertical availability for each station; areal availability for each stratum; area fished (for pot surveys); length-weight coefficients.

Commands defining the calculation (Section 3.1.4). User's preference for various options in the calculations: how should the distance towed at each station be calculated (from start and finish positions, or from recorded speed and time, etc.); what should be the width swept at each station (the recorded doorspread or a specified constant); what sub-populations, if any, should biomass be calculated for; how should LFs be scaled; information needed for phase-2 calculations and projected c.v.s.

Commands defining output (Section 3.1.5). What tables should be included in the output file and what degree of precision should be used for different categories of output (fish density, biomass, LF numbers, c.v., gain)?

Some examples of main input files are given in Section 3.1.6. Commands that may be (and sometimes must be) repeated in the main input file are discussed in Section 3.1.7.

3.1.1 Commands defining the data

The commands in this section define what type of survey is to be analysed (trawl or pot), which trip(s) and species are to be analysed, whether the data are to be read from a database or flat files and, if the former, which stations to use from the specified trip(s).

If the data are to be read from a database, some of all of the data in Table 1 will be extracted. Which tables are extracted depends on the specific analysis.

Table 1: Database tables from which data may be extracted by SurvCalc, and the attributes extracted from each table. The database and table names given are those for the original implementation (on Empress at Greta Point); slightly different names might apply in other situations.

Database	Table	Attributes extracted
<i>trawl</i>	<i>t_stratum</i>	trip_code, stratum, area_km2
<i>trawl</i>	<i>t_station</i>	trip_code, station_no, stratum, distance, lat_s, NorS_s, long_s, EorW_s, lat_s, NorS_f, long_f, EorW_f, speed, dist_doors and any other attributes included in @input_from_database.t_station_columns
<i>trawl</i>	<i>t_catch</i>	trip_code, station_no, species, weight
<i>trawl</i>	<i>t_subcatch</i> ¹	trip_code, station_no, species, subcatch_no, weight
<i>trawl</i>	<i>t_lgth</i>	trip_code, station_no, species, subcatch_no ² , lgth, percent_samp, no_a, no_m, no_f
<i>rdp</i>	<i>lw_coeff</i>	spp_code, sex lw_coeff_a, lw_coeff_b, lw_coeff_c

¹This table extracted only for task `output_for_catch_at_age` (see Section 5.7)

²This attribute extracted only for task `output_for_catch_at_age` (see Section 5.7)

@survey_type	Type of survey
Type	String (must be either <code>trawl</code> or <code>pot</code>)
Default	<code>trawl</code>
Effects	Determines which alternative variables and equations will be used in calculations and what other input file commands are valid. Other types of survey that may be allowed in future versions of SurvCalc include <code>scallop</code> and <code>oyster</code> .
Notes	This command is not needed for trawl surveys.
@trips	The trip, or trips, that should be analysed
Type	String vector, each member of which must be a valid trip code
Effects	Limits the data extracted from the database to that relating to the specified trip or trips

@species	The species to be analysed for a specified trip
Label	A trip code (must be in the vector <code>trips</code>)
Effects	Defines any following subcommands as being <code>@species</code> subcommands for the specified trip
Notes	Omit the label if only one trip is being analysed (i.e., if <code>trips</code> is of length 1).
codes	Species codes
Command	<code>@species</code>
Conditions	Only one species can be analysed (i.e., <code>codes</code> must have length 1) if the task is <code>output_for_catch_at_age</code> .
Type	String vector, each member of which must be a valid species code
Effects	Limits the catch and/or length data extracted from the database for the specified trip to that relating to the specified species.
@input_from_database	The interface to the database to extract the data
Effects	Defines any following subcommands as being <code>@input_from_database</code>
Notes	Ignored if the user has provided flat file input data (with run-time arguments -S, -T, -U, -V, or -W)
database	database
Command	<code>@input_from_database</code>
Conditions	Either Empress or Postgresql
Type	String
Default	Empress
Effects	Specifies the database from which the data are to be extracted.
hostname	Postgresql server name
Command	<code>@input_from_database</code>
Conditions	Only used for Postgresql database, and only needed when this database is being accessed across a network (i.e., when SurvCalc is not running on the machine on which the Postgresql database resides)
Type	String
Effects	Specifies the machine name on which the Postgresql database resides.
schema	schema name
Command	<code>@input_from_database</code>
Conditions	Only used for Postgresql database
Type	String
Default	trawl
Effects	Specifies the Postgresql schema name under which data tables are stored.
database_name	database name
Command	<code>@input_from_database</code>
Type	String
Default	trawl for Empress or fish for Postgresql
Effects	Specifies the database name in which data tables are stored.
t_station_columns	the additional attributes to be extracted from t_station table
Command	<code>@input_from_database</code>
Type	String vector
Conditions	The attributes must exist in <code>t_station</code> table of the database
Effects	Specifies the additional attributes to be extracted from <code>t_station</code> table.

@where		Restrict the selects from the database tables
Label		A trip code (must be in the vector <code>trips</code>)
Effects		Restricts the records selected from one or more of database tables <code>t_stratum</code> , <code>t_station</code> , <code>t_catch</code> , and <code>t_lgth</code> . Defines any following subcommands as being <code>@where</code> subcommands for the specified trip.
Notes		Most users will want to use only subcommand <code>t_station</code> (to define the station select). The label may be omitted if only one trip is being analysed (i.e., if <code>trips</code> is of length 1).
Example		The following command block restricts the stations selected to those for which <code>gear_perf</code> is less than 3 and <code>station_no</code> is less than 100 @where <code>t_station gear_perf < 3 and station_no < 100</code>
t_station		Restrictions for extracting station data
Command		@where
Type		String (must be a valid SQL Boolean expression)
Effects		Specifies criteria to restrict the selection of station data from <code>t_station</code> table.
Notes		SurvCalc automatically constructs an SQL to extract data from table <code>t_station</code> , and this ends with a 'where' clause restricting this extraction to the specified trip(s). SurvCalc will append <code>t_station</code> to this 'where' clause using 'and'.
t_stratum		Restrictions for extracting stratum data
Command		@where
Type		String (must be a valid SQL Boolean expression)
Effects		Specifies criteria to restrict the selection stratum data from <code>t_stratum</code> table.
Notes		SurvCalc automatically constructs an SQL to extract data from table <code>t_stratum</code> , and this ends with a 'where' clause restricting this extraction to the specified trip(s). SurvCalc will append <code>t_stratum_where</code> to this 'where' clause using 'and'.
t_catch		Restrictions for extracting catch data
Command		@where
Type		String (must be a valid SQL Boolean expression)
Effects		Specifies criteria to restrict the selection catch data from <code>t_catch</code> table.
Notes		SurvCalc automatically constructs an SQL to extract data from table <code>t_catch</code> , and this ends with a 'where' clause restricting this extraction to the specified trip(s) and species. SurvCalc will append <code>t_catch_where</code> to this 'where' clause using 'and'.
t_lgth		Restrictions for extracting length data
Command		@where
Type		String (must be a valid SQL Boolean expression)
Effects		Specifies criteria to restrict the selection catch data from <code>t_lgth</code> table.
Notes		SurvCalc automatically constructs an SQL to extract data from table <code>t_lgth</code> , and this ends with a 'where' clause restricting this extraction to the specified trip(s) and species. SurvCalc will append <code>t_lgth_where</code> to this 'where' clause using 'and'.

3.1.2 Commands modifying the data

The commands in this section allow the user to modify the stratification in the data that have been read in (either from a database or flat files). The following changes can be made: stations can be reassigned to different strata (either existing ones, or new user-defined strata); and areas of existing strata can be changed.

@change_strata	Reassign all stations in some strata to different strata (new or existing)
Label	A trip code (must be in the vector <code>trips</code>)
Conditions	It is a fatal error if there is any overlap between the stations affected by a <code>@change_strata</code> command and a <code>@reassign_strata</code> command for the same trip.
Effects	Defines any following subcommands as being <code>@change_strata</code> subcommands for the specified trip. All stations from the specified trip that were originally assigned to one of the strata listed in <code>to</code> are reassigned to the corresponding stratum in <code>from</code>
Notes	Omit the label if only one trip is being analysed (i.e., if <code>trips</code> is of length 1).
Examples	The following example assigns all stations in stratum 0023 or 0025 to stratum 023A and all stations in stratum 0027 to stratum 0030 <pre>@change_strata from 0023 0025 0027 to 023A 023A 0030</pre>
from	Names of strata whose stations are to be reassigned
Command	<code>@change_strata</code>
Conditions	Each string in <code>from</code> must be an existing stratum name for the specified trip
Type	String vector
to	Names of strata to which stations are to be reassigned
Command	<code>@change_strata</code>
Conditions	Each string in <code>to</code> must be either an existing stratum or defined in command <code>@new_strata</code> (i.e., must be in <code>new_strata[trip].names</code>)
Type	String vector of same length as <code>from</code>
@reassign_strata	Reassign some stations to different strata (new or existing)
Label	A trip code (must be in the vector <code>trips</code>)
Conditions	It is a fatal error if there is any overlap between the stations affected by a <code>@change_strata</code> command and a <code>@reassign_strata</code> command for the same trip.
Effects	Defines any following subcommands as being <code>@reassign_strata</code> subcommands for the specified trip. Each station in <code>stations</code> is reassigned to the corresponding stratum in <code>strata</code>
Notes	Omit the label if only one trip is being analysed (i.e., if <code>trips</code> is of length 1).
Examples	In the following example station 23 is reassigned to stratum 0012 and station 37 to stratum 012A <pre>@reassign strata tan0601 stations 23 37 strata 0012 012A</pre>
stations	Numbers of those stations which are to be reassigned to different strata
Command	<code>@reassign_strata</code>
Conditions	Each number in <code>stations</code> must be an existing station number for the specified trip
Type	Integer vector

strata	Names of the strata to which stations are to be reassigned
Command	@reassign_strata
Conditions	Each string in strata must be an existing stratum name for the specified trip or must be defined in command @new_strata (i.e., must be in new_strata[trip].names)
Type	String vector
@new_strata	Define new strata
Label	A trip code (must be in the vector trips)
Effects	Defines any following subcommands as being @new_strata subcommands for the specified trip.
Notes	Omit the label if only one trip is being analysed (i.e., if trips is of length 1). Areal availabilities for the new strata will be assumed to be 1. Different values, which are trip- and species-specific, may be set using command @areal_availability
Example	The following command creates new strata 003A and 003B for trip tan0601 with areas 2153 and 397, respectively, and areal-availabilities 1 and 0.8, respectively <pre>@new_strata tan0601 strata 003A 003B areas 2153 397</pre>
strata	Names of new strata
Command	@new_strata
Conditions	Must be different from the names of existing strata
Type	String vector
Effects	Defines the names of new strata
Notes	A warning should be output if any string in names does not occur in either change_strata[trip].to or reassign_strata[trip].strata
areas	Area (km²) of each new stratum
Command	@new_strata
Type	A numeric vector of the same length as names
Effects	Defines the areas of new strata
Notes	
@change_stratum_area	Change the areas of existing strata
Label	A trip code (must be in the vector trips)
Effects	Defines any following subcommands as being @change_stratum_area subcommands for the specified trip. Changes the area of the strata with names in names to the values in new_areas
Notes	Omit the label if only one trip is being analysed (i.e., if trips is of length 1).
Example	The following command block changes the area of strata 004A and 004B to 3152 and 793, respectively <pre>@change_stratum_area tan0601 strata 004A 004B new_areas 3152 793</pre>
strata	Names of strata whose areas are to be changed
Command	@change_stratum_area
Conditions	Must be names of existing strata
Type	String vector
new_areas	New areas for strata whose areas are to be changed
Command	@change_stratum_area
Type	Numerical vector

3.1.3 Commands extending the data

The commands in this section allow the user to extend the data to be analysed by providing length-weight coefficients or setting the multiplicative factors which affect either the calculation of fish density (vulnerability, vertical availability, and area fished – see Section 5.2) or the calculation of biomass from fish density (areal availability or population area – see Section 5.3). Length-weight coefficients need be provided only if they are needed (see Section 5.1.6) and if the default values in the database are not present or correct. Most of the multiplicative factors have default values (1 for vulnerability and both areal and vertical availability; the stratum area for population area). Other values must be provided separately for each combination of trip and species. Note that vulnerability, vertical availability, and area fished are associated with stations; areal availability and population area are associated with strata.

@vulnerability	Vulnerability of a species to capture at each station in a trip
Label	[trip_code]_[species_code]
Conditions	Ignored if the user has provided station-catch data (-W). Must not be used when <code>survey_type = pot</code>
Effects	Defines any following subcommands as being <code>@vulnerability</code> subcommands. Specifies a vulnerability for the given species at all stations in the given trip.
Notes	Each <code>@vulnerability</code> command block applies to one species in one trip. The command may be omitted when the vulnerability of the species is 1 at all stations in the trip.
Examples	The following command block specifies that the vulnerability of HOK in trip tan0601 is 0.9 and 0.8 for stations 23 and 25, respectively, and 1 for all other stations. <pre>@vulnerability tan0601_HOK default_value 1 other_stations 23 25 other_values 0.9 0.8</pre>
default_value	Default value for vulnerability of the given species in the given trip
Command	<code>@vulnerability</code>
Type	Positive number
Default	1
Effects	Defines the vulnerability of the given species in all stations in the given trip except for those in <code>other_stations</code>
other_stations	Stations at which the vulnerability differs from the default value
Command	<code>@vulnerability</code>
Conditions	Must be existing stations
Type	Numeric vector
Effects	Specifies stations at which the vulnerability differs from the default value
other_values	Vulnerability values that differ from the default value
Command	<code>@vulnerability</code>
Type	Positive numeric vector of same length as <code>other_stations</code>
Effects	Specifies the vulnerabilities for those stations in <code>other_stations</code>

@vertical_availability	Vertical availability of a species at each station in a trip
Label	[trip_code]_[species_code]
Conditions	Ignored if the user has provided station-catch data (-W). Must not be used when survey_type = pot.
Effects	Defines any following subcommands as being @vertical_availability subcommands. Specifies a vertical availability for the given species at all existing stations in the given trip.
Notes	Each @vertical_availability command block applies to one species in one trip. The command may be omitted when the vertical availability of the species is 1 at all stations in the trip.
Examples	The following command block specifies that the vertical availability of HOK in trip tan0601 is 0.8 and 1.2 for stations 33 and 35, respectively, and 1 for all other stations. @vertical_availability tan0601_HOK default_value 1 other_stations 33 35 other_values 0.8 1.2
default_value	Default value for vertical availability of the given species in the given trip
Command	@vertical_availability
Type	Positive number
Default	1
Effects	Defines the vertical availability of the given species in all stations in the given trip except for those in other_stations
other_stations	Stations at which the vertical availability differs from the default value
Command	@vertical_availability
Conditions	Must be existing stations for the given trip
Type	Numeric vector
Effects	Specifies stations at which the vertical availability differs from the default value
other_values	Vertical availability values that differ from the default value
Command	@vertical_availability
Type	Positive numeric vector of same length as other_stations
Effects	Specifies the vertical availabilities for those stations in other_stations
@area_fished	Area fished (m²) for a species at each station in a potting survey
Label	[trip_code]_[species_code]
Conditions	Must not be used except when survey_type = pot.
Effects	Defines any following subcommands as being @area_fished subcommands. Specifies an area fished for the given species at all existing stations in the given trip.
Notes	Each @area_fished command block applies to one species in one trip.
Examples	The following command block specifies that the area fished for BCO in trip abc0601 is 27 m ² and 25 m ² for stations 33 and 35, respectively, and 30 m ² for all other stations. @area_fished abc0601_BCO default_value 30 other_stations 33 35 other_values 27 25
default_value	Default value for area fished for the given species in the given trip
Command	@area_fished
Type	Positive number
Default	None
Effects	Defines the area fished for the given species at all stations in the given trip except for those in other_stations

other_stations	Stations at which the area fished differs from the default value
Command	@area_fished
Conditions	Must be existing stations for the given trip
Type	Numeric vector
Effects	Specifies stations at which the area fished differs from the default value
other_values	Area fished values that differ from the default value
Command	@area_fished
Type	Positive numeric vector of same length as other_stations
Effects	Specifies the areas fished for those stations in other_stations
@areal_availability	Areal availability of a species at each stratum in a trip
Label	[trip_code]_[species_code]
Conditions	For each combination of trip and species to be analysed there must not be both an @areal_availability command block and a @population_area command block (use one or the other, or neither).
Effects	Defines any following subcommands as being @areal_availability subcommands. Specifies an areal availability for the given species at all existing strata in the given trip.
Notes	Each @areal_availability command block applies to one species in one trip. The command may be omitted when the areal availability of the species is 1 at all strata in the trip.
Examples	The following command block specifies that the areal availability of HOK in trip tan0601 is 0.8 and 1.2 for strata 0003 and 0004, respectively, and 1 for all other strata. @areal_availability tan0601_HOK default_value 1 other_strata 0003 0004 other_values 0.8 1.2
default_value	Default value for areal availability of the given species in the given trip
Command	@areal_availability
Type	Positive number
Default	1
Effects	Defines the areal availability of the given species in all strata in the given trip except for those in other_strata
other_strata	Strata in which the areal availability differs from the default value
Command	@areal_availability
Conditions	Must be existing strata for the given trip
Type	String
Effects	Specifies strata in which the areal availability differs from the default value
other_values	Areal availability values that differ from the default value
Command	@areal_availability
Type	Positive numeric vector of same length as other_strata
Effects	Specifies the areal availabilities for those strata in other_strata

@population_area	Population area for species in each stratum in a trip
Label	[trip_code]_[species_code]
Conditions	For each combination of trip and species to be analysed there must not be both an @areal_availability command block and a @population_area command block (use one or the other, or neither).
Effects	Defines any following subcommands as being @population_area subcommands. Specifies a population area for the given species at all existing strata in the given trip. This is used to calculate areal availability, which is population area divided by stratum area
Notes	Each @population_area command block applies to one species in one trip. The command may be omitted when the population area of the species is the same as the stratum area for all strata in the trip.
Examples	The following command block specifies that the population area of HOK in trip tan0601 is 2957 and 1325 for strata 0003 and 0004, respectively, and equal to the stratum area for all other strata. @population_area other_strata 0003 0004 other_values 2957 1325
other_strata	Strata in which the population area differs from the stratum area
Command	@population_area
Conditions	Must be existing strata for the given trip
Type	String
Effects	Specifies strata in which the population area differs from the stratum area
other_values	Population areas that differ from stratum area
Command	@population_area
Type	Positive numeric vector of same length as other_strata
Effects	Specifies the population area for those strata in other_strata
@lw_coeff	Length-weight coefficients for a species in a trip
Label	[trip_code]_[species_code]
Conditions	Needed only if length-weight coefficients are required for an analysis and the user wants to use values different from those in database rdb. If this command is used it must be repeated for each combination of trip and species for which length-weight coefficients are required (see Section 5.1.6). If it is not used, then all required length-weight coefficients will be read from rdb.
Effects	Defines any following subcommands as being @lw_coeff subcommands. Specifies the coefficients used to calculate the weight of a fish (in g) from its length (in cm).
Notes	Note that length-weight coefficients are not always required in analyses (see Section 5.1.6).
Examples	The following command block specifies the length-weight coefficients (independent of sex) for HOK in trip tan0601. @lw_coeff tan0601_HOK a 0.006 b 2.85
a, b, c	Length-weight coefficients [to calculate the weight of a fish (in g) from its length (in cm)].
Command	@lw_coeff
Conditions	Use either a, b, and (optionally) c, or a_male, b_male, a_female, b_female, and (optionally) c_male, c_female, a_unsexed, b_unsexed, c_unsexed
Type	Numeric
Default	c = 1; no default for a or b

a_male, b_male, c_male, a_female, b_female, c_female Length-weight coefficients by sex [to calculate the weight of a fish (in g) from its length (in cm)].

Command	@lw_coeff
Conditions	Use either a, b, and (optionally) c, or a_male, b_male, a_female, b_female, and (optionally) c_male, c_female, a_unsexed, b_unsexed, c_unsexed
Type	Numeric
Default	c_male = 1 and c_female = 1; no default for other coefficients

a_unsexed, b_unsexed, c_unsexed Length-weight coefficients for unsexed [to calculate the weight of a fish (in g) from its length (in cm)].

Command	@lw_coeff
Conditions	Use either a, b, and (optionally) c, or a_male, b_male, a_female, b_female, and (optionally) c_male, c_female, a_unsexed, b_unsexed, c_unsexed
Type	Numeric
Default	c_unsexed = 1; no default for other coefficients
Notes	If length-weight coefficients are presented for males and females, but not for unsexed fish, then the weight of an unsexed fish of a given length is calculated as the average of the weights of a male and female of that length.

3.1.4 Commands defining the calculations

The commands in this section describe the user's preference for various options in the calculations. There are options associated with fish density (see Section 5.1.4), sub-populations defined by sex and/or length ranges (see Section 5.3.1), and the scaling of LFs (see Section 5.4).

@preferences	User preferences for fish density calculations
Label	A trip code (must be in the vector <code>trips</code>)
Conditions	Ignored if: the task is <code>output_lw_coeff</code> ; or if it is <code>calc_LFs</code> and <code>LF_scaling</code> is not <code>numbers_per_km2</code> ; or if the user has provided station-catch data (-W).
Effects	Defines any following subcommands as being <code>@preferences</code> subcommands for the specified trip.
Notes	Omit the label if only one trip is being analysed (i.e., if <code>trips</code> is of length 1).
distance_towed	User preference for calculation of distance towed at each station
Command	<code>@preferences</code>
Conditions	Must not be used when <code>survey_type = pot</code>
Type	String vector containing one or more of the following options in order of preference: <code>recorded_distance</code> , <code>recorded_speed*time</code> , <code>constant_speed*time</code> , <code>from_lat_lon</code>
Default	None
Effects	Defines the user preference for the method (or methods, in priority order) of defining the distance towed at each station. Any station for which distance towed can not be defined will be dropped from the analysis.
Notes	See Section 5.1.4 for more details about the options for <code>distance_towed</code>
width_swept	User preference for defining width swept at each station
Command	<code>@preferences</code>
Conditions	Must not be used when <code>survey_type = pot</code>
Type	String vector containing one or more of the following options in order of preference: <code>recorded_doorspread</code> , <code>constant_doorspread</code> , <code>recorded_wingspread</code>
Default	None
Effects	Defines the user preference for the method (or methods, in priority order) of defining the width swept at each station. Any station for which width swept can not be defined will be dropped from the analysis.
Notes	See Section 5.1.4 for more details about the options for <code>width_swept</code>
catch_weight	User preference for calculation of catch weight at each station
Command	<code>@preferences</code>
Type	String vector containing one or more of the following options in order of preference: <code>recorded</code> , <code>calculated</code>
Default	None
Effects	Defines the user preference for the method (or methods, in priority order) of defining the catch weight at each station. Any station for which catch weight can not be defined will be dropped from the analysis.
Notes	See Section 5.1.4 for more details about the options for <code>catch_weight</code>

@sub_populations	sub-populations for which biomasses should be calculated
Label	A species code (must be in <code>species[trip].code</code> for some trip in <code>trips</code>)
Conditions	Ignored unless the task is one of <code>calc_biomass</code> , <code>calc_biomass_and_LFs</code> , or <code>phase_2_calc</code> .
Effects	Defines any following subcommands as being <code>@sub_populations</code> subcommands for the specified species.
Notes	For analyses covering more than one trip this command will affect the output for the given species in all trips where that species is analysed.
Examples	The following command requests ORH biomass estimates for six sub-populations, and supplies labels to be used for these in the output file: <pre>@sub_populations ORH sexes all all male male female female Lmin 0 20 0 20 0 20 Lmax 20 80 20 80 20 80 labels A_sm A_lg M_sm M_lg F_sm F_lg</pre>
sexes	Sexes of sex-based sub-populations
Command	<code>@sub_populations</code>
Type	String vector containing one or more of <code>all</code> , <code>male</code> , <code>female</code> , and <code>unsexed</code>
Effects	Specifies the sex of each sub-population for the given species
Lmin, Lmax	Length bounds for each length-based sub-population
Command	<code>@sub_populations</code>
Type	Numeric vector of same length as <code>sexes</code>
Effects	Specifies length bounds for each sub-population for the given species
Notes	Sub-population includes all fish with $L_{min} \leq \text{length} < L_{max}$
labels	User-supplied labels for sex-based sub-populations
Command	<code>@sub_populations</code>
Type	String vector of same length as <code>sexes</code>
Effects	Supplies labels for sub-population biomass estimates in the output
@constant_speed	Vessel speed to be used in calculating distance-towed
Label	A trip code (must be in the vector <code>trips</code>)
Conditions	Ignored if the user has provided station-catch data (-W). This command must be provided for all trips in which <code>constant_speed*time</code> occurs in <code>preferences[trip]distance_towed</code> . It will be ignored for other trips.
Effects	Defines any following subcommands as being <code>@constant_speed</code> subcommands for the specified trip.
Notes	Omit the label if only one trip is being analysed (i.e., if <code>trips</code> is of length 1).
value	User-supplied vessel speed for the given trip
Command	<code>@constant_speed</code>
Type	Number
Notes	This vessel speed will be used only for those stations in the specified trip in which <code>constant_speed*time</code> is the selected option for calculating distance towed (as specified in command <code>@preferences</code>)

@constant_doorspread Doorspread to be used in calculating the width swept	
Label	A trip code (must be in the vector <code>trips</code>)
Conditions	Ignored if the user has provided station-catch data (-W). This command must be provided for all trips in which <code>constant_doorspread</code> occurs in <code>preferences[trip]width_swept</code> . It will be ignored for other trips.
Effects	Defines any following subcommands as being <code>@constant_doorspread</code> subcommands for the specified trip.
Notes	Omit the label if only one trip is being analysed (i.e., if <code>trips</code> is of length 1).
value	User-supplied vessel doorspread for the given trip
Command	<code>@constant_doorspread</code>
Type	Number
Notes	This doorspread will be used only for those stations in the specified trip in which <code>constant_doorspread</code> is the selected option for calculating width swept (as specified in command <code>@preferences</code>)
@LF_scaling Method of scaling LFs	
Conditions	Ignored unless LFs are being calculated (i.e., the task is <code>calc_biomass_and_LFs</code> or <code>calc_LFs</code>). Option <code>numbers_in_population</code> is obligatory if the task is <code>calc_biomass_and_LFs</code> , and not allowed if it is <code>calc_LFs</code> .
Type	String (must be one of <code>unscaled</code> , <code>numbers_per_1000</code> , <code>numbers_per_tow</code> , <code>numbers_per_hour</code> , <code>numbers_per_km2</code> , <code>numbers_in_population</code>)
Default	<code>numbers_in_population</code> in <code>-B</code> run, no default in <code>-l</code> run
Effects	Defines how LFs are scaled (see Section 5.4)
Notes	This command applies to all trips and species analysed. It can be omitted if the task is <code>calc_biomass</code> , in which case there is only one <code>LF_scaling</code> option.
@phase_2 User preferences for phase-2 gain calculations	
Label	A species code (must be in <code>species[trip].code</code>)
Conditions	Ignored unless the task is <code>phase_2_calc</code> .
Effects	Defines any following subcommands as being <code>@phase_2</code> subcommands.
Notes	If this command is used after some phase-2 stations have already been carried out then you should use command <code>@where</code> to exclude these stations from this analysis. It is not necessary to specify a trip for the phase-2 calculations because it is a fatal error to request <code>phase_2_calc</code> when <code>trips</code> is of length > 1.
algorithm Algorithm to be used in calculating gains	
Command	<code>@phase_2</code>
Type	String (must be either <code>mean_squared</code> or <code>variance</code>).
Default	<code>mean_squared</code>
Effects	Defines which algorithm will be used in calculating gain (see Section 5.5).
Notes	This sub-command is included only for completeness. Users should omit it unless they have a very good reason to use the variance algorithm.
n_stations Maximum number of additional stations per stratum	
Command	<code>@phase_2</code>
Type	Positive integer
Default	None
Effects	Defines the maximum number of additional stations in each stratum for which gains will be calculated (i.e., the number of columns in the output table of gains – see Section 5.5).

@projected_cvs	Calculate projected biomass c.v.s during a survey
Conditions	Ignored unless <code>run_time</code> task is <code>calc_biomass</code> or <code>calc_biomass_and_LFs</code> . Can only be used in a single-trip analysis.
Effects	Calculates the projected biomass c.v.s (i.e., the c.v.s expected at the end of the survey, given the data to date – see Section 5.6). Defines any following subcommands as being <code>@projected_cvs</code> subcommands.
Example	The following command block could be used during a 2-phase survey at the time when all stations had been completed except for 2 in stratum 002A and 3 in stratum 002B. <pre>@projected_cvs strata 002A 002B extra_stations 2 3</pre>
strata	Strata that are not yet completed
Command	<code>@projected_cvs</code>
Conditions	There must already be at least two stations in all strata specified here
Type	String vector of stratum names
Effects	Specifies which strata are not yet completed (i.e., those in which there are still some more stations to be done)
extra_stations	Number of additional stations per stratum
Command	<code>@projected_cvs</code>
Type	Constant vector of same length as <code>strata</code>
Effects	Specifies how many more stations are to be done in each stratum named in subcommand <code>strata</code>

3.1.5 Commands defining output

The main output file produced by SurvCalc contains obligatory and optional elements. The commands in this section, together with command `@station_data_columns` in Section 3.1.1, allow the user to specify which optional elements should be included. They also specify to what precision certain numbers (in both the main and other output files) should be given.

@output_tables	Tables to include in the main output file
Label	None
Conditions	Ignored unless biomass or LFs are being calculated (i.e., the task is <code>calc_biomass</code> , <code>calc_biomass_and_LFs</code> , or <code>calc_LFs</code>)
Effects	Defines any following subcommands as being <code>@output_tables</code> subcommands.
Notes	See Section 4.1.1 for descriptions of all obligatory and optional tables of biomass and LFs.
Examples	The following example requests tables of sub-population biomass and LFs by stratum, but not LFs by station. <pre>@output_tables sub_biomass_by_stratum T LFs_by_stratum T LFs_by_station F</pre>

sub_biomass_by_stratum **Output table(s) of sub-population biomass estimates (and c.v.s) by stratum**

Command	<code>@output_tables</code>
Conditions	Ignored unless biomass is being calculated (i.e., the task is <code>calc_biomass</code> or <code>calc_biomass_and_LFs</code>).
Type	Logical
Default	False
Notes	A separate table is output for each combination of species and trip (but excluding species for which no sub-populations have been defined).

biomass_by_species **Output table(s) of all population biomass estimates from a trip by species and stratum**

Command	<code>@output_tables</code>
Conditions	Ignored unless biomass is being calculated (i.e., the task is <code>calc_biomass</code> or <code>calc_biomass_and_LFs</code>).
Type	Logical
Default	False
Notes	A separate table is output for each trip in which more than one species is analysed, and this includes only those species in <code>species[trip].codes</code> . This table does not include estimates of sub-population biomass.

biomass_by_species_stratum **Output a table of all population biomass estimates by species and stratum**

Command	<code>@output_tables</code>
Conditions	Ignored unless biomass is being calculated (i.e., the task is <code>calc_biomass</code> or <code>calc_biomass_and_LFs</code>).
Type	Logical
Default	False
Notes	A separate table is output for each trip in which more than one species is analysed, and this includes only those species in <code>species[trip].codes</code> . This table does not include estimates of sub-population biomass.

biomass_by_species_trip	Output a table of all population and sub-population biomass estimates by species and trip
Command	@output_tables
Conditions	Ignored unless biomass is being calculated (i.e., the task is <code>calc_biomass</code> or <code>calc_biomass_and_LFs</code>) and more than one trip is being analysed.
Type	Logical
Default	False
Notes	A single table is output, after all other outputs, which combines the information in the biomass tables for each species-trip combination.
LFs_by_stratum	Output table(s) of LFs by stratum
Command	@output_tables
Conditions	Ignored unless LFs are being calculated (i.e., the task is <code>calc_LFs</code> or <code>calc_biomass_and_LFs</code>).
Type	Logical
Default	False
Notes	A separate table is output for each combination of species and trip.
LFs_by_station	Output table(s) of LFs by station
Command	@output_tables
Conditions	Ignored unless LFs are being calculated (i.e., the task is <code>calc_LFs</code> or <code>calc_biomass_and_LFs</code>).
Type	Logical
Default	False
Notes	A separate table is output for each combination of species and trip.
Number_measured	Output table(s) of numbers of fish measured by stratum and sex
Command	@output_tables
Conditions	Ignored unless LFs are being calculated (i.e., the task is <code>calc_LFs</code> or <code>calc_biomass_and_LFs</code>).
Type	Logical
Default	False
Notes	A separate table is output for each combination of species and trip.
LF_totals	Output table(s) of LF totals by stratum
Command	@output_tables
Conditions	Ignored unless LFs are being calculated (i.e., the task is <code>calc_LFs</code> or <code>calc_biomass_and_LFs</code>).
Type	Logical
Default	False
Notes	A separate table is output for each combination of species and trip.

@output_precision	Precision of quantities in the output files
Label	None
Conditions	Ignored when the task is <code>output_LW_coeffs</code> .
Effects	Defines any following subcommands as being <code>@output_precision</code> subcommands.
Notes	Use this command only if the default level of precision, as given in the following example, is unsatisfactory. This command affects both the main output file and files <code>station.out</code> , <code>station_catch.out</code> , and <code>stratum_catch.out</code> (see Section 4.4 for more details).
Examples	The following example shows the default level and type of precision for all output quantities. <pre>@output_precision quantity density biomass LF_number cv gain type sig_fig sig_fig sig_fig dec_place dec_place precision 4 4 4 1 0</pre>
quantity	Type of quantity for which precision is being specified
Command	<code>@output_precision</code>
Type	String vector containing any or all of <code>density</code> , <code>biomass</code> , <code>LF_number</code> , <code>cv</code> , <code>gain</code>
type	Type of precision being specified (significant figures or decimal places)
Command	<code>@output_precision</code>
Type	String vector of same length as <code>quantity</code> containing either <code>sig_fig</code> or <code>dec_place</code> at each position
precision	Number of significant figures or decimal places for each quantity
Command	<code>@output_precision</code>
Type	Integer vector of same length as <code>quantity</code>

3.1.6 Examples of main input files

One simple way of creating a valid main input file is to edit an existing file. With that in mind, here are three example files of increasing complexity. All could be used with tasks `calc_biomass` or `calc_biomass_and_LFs`. They could also be used with task `calc_LFs` if the LF scaling was specified (by adding a command like `@LF_scaling numbers_per_hour`).

Example 1

The first example is for the simplest analysis: for one species in one trip (the `@vulnerability` command in this file won't usually be needed but is included in all three examples because it illustrates a couple of useful points).

```
@trips tan0617

@species tan0617
codes HOK

@input_from_database
database Empress

@where
t_station gear_perf < 3 and station_no != 60

@preferences
distance_towed recorded_distance recorded_speed*time from_lat_long
width_swept recorded_doorspread
catch_weight recorded calculated
```

```
@vulnerability tan0617_HOK
default_value 1
other_stations 71 72
other_values 0.8 0.8
```

```
@sub_populations
sexes all all
Lmin 0 60
Lmax 59 200
labels to60 60+
```

```
@output_tables
sub_biomass_by_stratum T
biomass_by_species F
biomass_by_species_stratum F
LFs_by_stratum T
LFs_by_station F
Number_measured T
LF_totals T
```

```
@output_precision
quantity density biomass LF_number cv gain
type dec_place dec_place sig_fig dec_place dec_place
precision 0 0 8 0 1
```

Example 2

The second example extends the first to analyse multiple species in one trip. It differs from the first example in only two places. The first, at subcommand `codes` of command `@species` specifies the additional species to be analysed. The second is required because the command `@sub_populations` requires a label (a species code) when multiple species are analysed. No sub-population biomasses will be calculated for the other species (HAK and LIN) unless an additional `@sub_populations` command block (or blocks) is included (with the appropriate label). Note also that, because of its label, the `@vulnerability` command block has no effect on the biomass estimates for HAK and LIN.

```
@trips tan0617
```

```
@species tan0617
codes HOK HAK LIN
```

```
@input_from_database
database Empress
```

```
@where
t_station gear_perf < 3 and station_no != 60
```

```
@preferences
distance_towed recorded_distance recorded_speed*time from_lat_long
width_swept recorded_doorspread
catch_weight recorded calculated
```

```
@vulnerability tan0617_HOK
default_value 1
other_stations 71 72
other_values 0.8 0.8
```

```

@sub_populations HOK
sexes all all
Lmin 0 60
Lmax 59 200
labels to60 60+

@output_tables
sub_biomass_by_stratum T
biomass_by_species F
biomass_by_species_stratum F
LFs_by_stratum T
LFs_by_station F
Number_measured T
LF_totals T

@output_precision
quantity density biomass LF_number cv gain
type dec_place dec_place sig_fig dec_place dec_place
precision 0 0 8 0 1

```

Example 3

The final example extends the second example to analyse multiple trips. Note that for each trip a list of species to be analysed, a `@where` command, and a set of preferences is provided. The `@sub_populations` command for HOK will apply to all trips for which this species is to be analysed.

```

@trips tan0617 tan0714

@species tan0617
codes HOK HAK LIN

@species tan0714
codes HOK HAK

@input_from_database
database Empress

@where tan0617
t_station gear_perf < 3 and station_no != 60

@where tan0714
t_station gear_perf < 3

@preferences tan0617
distance_towed recorded_distance recorded_speed*time from_lat_long
width_swept recorded_doorspread
catch_weight recorded calculated

@preferences tan0714
distance_towed recorded_distance recorded_speed*time from_lat_long
width_swept recorded_doorspread
catch_weight recorded

@vulnerability tan0617_HOK
default_value 1
other_stations 71 72
other_values 0.8 0.8

```

```

@sub_populations HOK
sexes all all
Lmin 0 60
Lmax 59 200
labels to60 60+

```

```

@output_tables
sub_biomass_by_stratum T
biomass_by_species F
biomass_by_species_stratum F
LFs_by_stratum T
LFs_by_station F
Number_measured T
LF_totals T

```

```

@output_precision
quantity density biomass LF_number cv gain
type dec_place dec_place sig_fig dec_place dec_place
precision 0 0 8 0 1

```

3.1.7 Repeated commands in the main input file

The examples in Section 3.1.6 have shown that some commands can be repeated within the main input file. When they are repeated, their labels show exactly what each command applies to (either a trip, a species, or a combination of a trip and species). Some of these commands must be repeated (e.g., @preferences must be repeated for each trip) and others need not be because there is a default action when they are not repeated (e.g., when @vulnerability is not repeated for a trip-species it is assumed to be 1 for all stations for that trip-species). The requirements for all commands that may be repeated are summarised in Table 2.

Labels may be omitted from these commands when this causes no ambiguity. For example, those commands that are usually labelled by trip need no label (a) when they are intended to apply to all trips, or (b) in a main input file that concerns only one trip. Similarly, if the file concerns only one trip and one species then the commands normally labelled by trip and species do not need a label.

Table 2: Requirements for all commands that can be repeated in a main input file.

Command	Label	Must be repeated?	Default action if command not repeated
@species	trip	Yes	
@preferences	trip	Yes	
@where	trip	No	Use default selects
@change_strata	trip	No	No changes
@reassign_strata	trip	No	No changes
@new_strata	trip	No	No new strata
@change_stratum_area	trip	No	No changes
@constant_speed	trip	Maybe ¹	
@constant_doorspread	trip	Maybe ¹	
@sub_populations	species	No	No sub-populations
@phase_2	species	Yes	
@vulnerability	trip & species	No	All = 1
@vertical_availability	trip & species	No	All = 1
@areal_availability	trip & species	No	All = 1
@population_area,	trip & species	No	All = stratum area
@area_fished	trip & species	Yes	
@lw_coeff	species ³ or trip & species	No ²	

¹Depends on @preferences; ²Only needed for trip-species combinations where length-weight coefficients are required (see Section 5.1.6); ³Use species label when the same coefficients are to be used for all trips.

3.2 Other input files

All input files other than the main one (see Section 3.1) are flat files. That is, files containing columns of data, with the column names in the first line. Columns should be separated by white space (i.e., spaces or tabs) and the order of the columns doesn't matter. Missing values should be entered as NULL. The user specifies which flat files (if any) SurvCalc should read using the run-time arguments -S, -T, -U, -V, -W, or -X (see Section 2). Each row of the flat file corresponds to: a stratum (with -S); or a station (with -T or -X); or a catch record (i.e., for each station there must be a row for each species caught at that station) (with -U); or a subcatch record (i.e., for each station there must be a row for each subcatch of species caught at that station) (with -W); or a length record (i.e., for each station, species, and subcatch, there must be a row for each length that occurs in the LF in that subcatch of that species at that station) (with -V).

The columns, and column names, required for the stratum (-S), catch (-U), subcatch file (-W), and length (-V) files are precisely those given in Table 1. For the station file (-T), the only obligatory columns are station_no and stratum (and trip_code if there is more than one trip). Which of the other columns from this table are required depends on command @preferences (e.g., column dist_doors is not needed if preferences.width_swept is constant_doorspread).

For the station-catch file (-X), the only obligatory columns are station_no, stratum, and a fish density with a label like XYZ_kg_km2, where XYZ is a user-specified species code (which need not be an official species code). There can be densities for more than one species (e.g., HOK_kg_km2, HAK_kg_km2). The only other optional column is trip_code (a user-provided code, which need not be an official trip code), which is required only if the user wishes to analyse more than one survey. Note that vulnerabilities and vertical availabilities can not be entered as columns in this file. When a station-catch file is used it is assumed that the fish density columns are already adjusted for vulnerability and vertical availability.

4. OUTPUT FILES

Most output from SurvCalc goes to the main output file (Section 4.1), which is named in the command line (e.g., `myfile` in the example command line at the beginning of Section 2). If requested, additional output goes to one or more flat files (Section 4.2) or a catch-at-age data file (Section 4.3). The precision of many numbers in the output files, as specified by command `@output_precision`, is discussed in Section 4.4.

4.1 Main output file

The structure of this file is most simply illustrated by the following, which shows the structure of the main output file expected when the input file is as in the last example in Section 3.1.6 and the task is `calc_biomass`, `calc_biomass_and_LFs`, or `calc_LFs` (the indenting in what follows is simply used to illustrate the structure of the file; it does not occur in the actual file).

Header

```
SELECTIONS for trip tan0617:
PREFERENCES for trip tan0617:
    Results for species HOK in trip tan0617
    Results for species HAK in trip tan0617
    Results for species LIN in trip tan0617
Tables for trip tan0617
SELECTIONS for trip tan0714:
PREFERENCES for trip tan0714:
    Results for species HOK in trip tan0714
    Results for species HAK in trip tan0714
Tables for trip tan0714
```

For tasks `output_flat_files` and `output_for_catch_at_age` the main output file contains only the header. For task `output_LW_coeffs`, it contains the header plus a table of length-weight coefficients.

The header contains general information about the particular run of SurvCalc, including the command line, the date, the version of SurvCalc (this is useful if bugs are subsequently found in the program), the user name, and, if the survey data have been extracted from a database, the SQLs that were used to do this. If LFs are calculated then the last line in the header describes the LF scaling.

The SELECTIONS section contains two lists: one of the selected strata (including their effective areas) and one of the selected stations (including the stratum associated with each). The PREFERENCES is simply a repeat of the information in command `@preferences`.

The results for a particular species and trip always starts with a list of exclusions (which stations and strata are excluded, and why) and summaries of the numbers of stations and strata that were selected and used. After this, the remaining output is in a series of tables, some obligatory and others optional, the latter being requested with command `@output_tables`.

4.1.1 Tables in the main output file

`area_swept_factors` Obligatory table when area swept is calculated; produced for each species-trip combination. Summary statistics for the two factors that define area swept.

	mean	min	max
<code>width_swept_m</code>	120.56	104	129.7
<code>distance_towed_km</code>	5.47	3.82	5.76

`stratum_summary` Obligatory table when biomasses are being calculated; produced for each species-trip combination. One line per stratum and including mean fish densities, biomass and c.v. `NtowsHOK` is the number of tows with non-zero catch for HOK (or whatever the specified species is). The area is the population area, a'_s (which may differ from the stratum area if commands `@areal_availability` or `@population_area` are used).

stratum	area	Ntows	NtowsHOK	kg/km2	biomass	cv.biomass
0001	2150	4	3	9	19	71
0002	1318	4	4	24	31	48
...						

`biomass` Obligatory table when biomasses are being calculated; produced for each species-trip combination. One line (the first, with label `all`) for the total biomass, and then one line for each requested sub-population, with user-supplied labels. Lower and upper bounds are estimate ± 2 s.e.s.

population	lbound	biomass	ubound	cv	Nstations
<code>all</code>	31601	46081	60561	16	97
<code>to60</code>	-114	846	1807	57	90
<code>60+</code>	11830	20001	28172	20	90
...					

`sub_biomass_by stratum` Optional table produced for each species-trip combination. Contains biomass estimates, and c.v.s, for a single species by sub-population and stratum; the column labels are user-supplied for each sub-population.

stratum	to60	cv_to60	60+	cv_60+
0001	726	66	639	85
0004	87	71	3345	50
...				

`biomass_by_species` Optional table produced for every trip in which more than one species is analysed. Like the biomass table above, containing one line (total biomass) per species.

species	lbound	biomass	ubound	cv	Nstations
HOK	31601	46081	60561	16	97
HAK	1827	2628	3429	15	97
...					

biomass_by_species_stratum Optional table produced for every trip in which more than one species is analysed. One line per stratum, containing biomass estimates, and c.v.s, for all species analysed for that trip.

```
stratum  HOK  cv_HOK  HAK  cv_HAK
0001     1365    74    19    71
0002     206    35    31    48
...
```

biomass_by_species_trip Optional table, presented at the end of the output from a multi-trip analysis, containing all biomass estimates, including those for sub-populations (but not stratum biomasses). This table is simply the union of all the biomass tables, ordered by species, and then by trip within species.

```
species  trip  population  lbound  biomass  ubound  cv  Nstations
HOK      tan0601      all    31601    46081    60561  16        97
HOK      tan0601      to60     -114      846     1807  57        90
HOK      tan0601      60+    11830    20001    28172  20        90
HOK      tan0701      all    29701    51315    63517  18        95
...
```

projected_cvs Obligatory table when biomasses are calculated and command @projected_cvs is used. Includes projected c.v.s for all total biomasses (including sub-populations, if any).

```
species  population  projected_cv
HOK      all        10
HOK      to60       23
HOK      60+       11
HAK      all        16
```

LF_correction_factors Obligatory table for task calc_biomass_and_LFs; produced for each species-trip combination (see CF_s and CF in Section 5.4).

```
stratum  correction_factor
0001          1.08
0002          1.08
...
all          1.04
```

LF_overall Obligatory table when LFs are calculated; produced for each species-trip combination. A single table for the whole population as follows (N means number measured; SN means scaled number).

```
L      Nfemale    Nmale  Nunsexed    Nall  SNfemale    SNmale  SNunsexed    SNall
32      1         3        0         4    2199.1    1783.3        0    3983.2
33      8         5        0        13    9259.1    16399.2        0   25658.3
...
```

LFs_by_stratum Optional table; produced for each species-trip combination. Same format as LF_overall, but with an additional initial column, stratum.

LFs_by_station Optional table; produced for each species-trip combination. Same format as LF_overall, but with an additional initial columns, stratum and station.

Number_measured Optional table; produced for each species-trip combination. One line per stratum (and a final summary line) containing the number of fish measured by sex and stratum.

stratum	Nfemale	Nmale	Nunsexed	Nall
0001	624	656	1	1281
0002	302	131	0	433
...				
All	2501	1765	9	4275

LF_totals Optional table; produced for each species-trip combination. One line per stratum (and a final summary line) containing LF totals, and their c.v.s, by stratum and sex

stratum	SNfemale	SNmale	SNunsexed	SNall	cv_female	cv_male	cv_unsexed	cv_all
0001	1163734.1	1463051.0	5269.5	2632054.6	39	44	100	35
0002	161545.6	69949.2	0	231494.9	17	8	0	36
...								
All	7732192.2	5210113.8	39712.7	12982019.0	12	18	70	12

gains Obligatory table for task phase_2_calc; produced for each species. One line per stratum, number of column determined by subcommand n_stations.

stratum	1	2	3	4	5	6	7	8	9	10
0001	16.8	11.2	8	6	4.7	3.7	3.1	2.5	2.2	1.8
0002	8.5	4.3	2.6	1.7	1.2	0.9	0.7	0.6	0.5	0.4
...										

4.2 Output to flat files

Flat files contain columns of data, with the column names in the first line. Each row of the flat file corresponds to: a stratum (with -s or -y); or a station (with -t or -w); or a catch record (with -u); or a subcatch record (with -w); or a length record (with -v).

For -s, -t, -u, -v, or -w the columns are the same as those extracted from the corresponding table in the database (see Section 3.1.1), or as provided by the user (with -S,-T,-U, -V, or -W).

4.2.1 Station-catch file

A file of station-catch data is output with -x switch in the command line if run time task is one of output_flat_files, calc_biomass, calc_LFs, or calc_biomass_and_LFs.

The file contains all the columns names in the station table plus three columns for each of the species analysed, as well as two columns for each of the sub_populations defined by the user. For species XYZ, these columns are XYZ_kg (the catch weight, C_i), XYZ_kg_km (the linear density, LD_i), and XYZ_kg_km2 (the areal density, AD_i). For sub_population ABC of species XYZ, the two columns are XYZ_ABC_kg (the catch weight, c_i), and XYZ_ABC_kg_km2 (the areal density, ad_i). For stations from trips in which species XYZ is not analyzed NULL is entered for all columns associated with this species.

4.2.2 Output to stratum-catch file

A file of stratum-catch data is output with -y switch in the command line if run time task is one of output_flat_files, calc_biomass, calc_LFs, or calc_biomass_and_LFs.

The file contains columns trip_code, stratum, and area_km2 plus three columns for each of the species analysed. For species XYZ, these columns are XYZ_kg (stratum mean catch weight, $(\sum_i C_i)/n_s$), XYZ_kg_km (the linear density, LD_s), and XYZ_kg_km2 (the areal density, AD_s). For strata from trips in which species XYZ is not analyzed NULL is entered for all columns associated with this species.

4.3 Catch-at-age data output

The command 'SurvCalc -c > myout' run produces, for every combination of trip and species in the input file, a file that can be read by the catch-at-age function import.length.data, as well as a 'main output file' ('myout' in the above example), whose format is described in Section 4.1. The catch-at-age files are either in the 'survey' or 'survey.sub' format, with the latter format being used if there are any subcatches for the given trip and species.

The names of the catch-at-age files are not specified by the user. They are constructed from the species and trip names in the form LFtttttccc.i (for format 'survey') or LFtttttccsub.i (for format 'survey.sub'), where ttttt is the trip code, ccc is the species code, and i is an index used by SurvCalc to ensure that any existing files don't get overwritten. For example, the first time you do a 'SurvCalc -c' run for HAK in trip tan0901 SurvCalc will produce the output file LFtan0901HOK.1 (or LFtan0901HOKsub.1 if there were any subcatches). If you then do another 'SurvCalc -c' run for the same species and trip, in the same directory without deleting the first output file, then the new output file will be LFtan0901HOK.2 (or LFtan0901HOKsub.2).

The 'survey' format is as follows:

```
comment line
trip
number_of_strata
length-weight parameters: male_a male_b female_a female_b {optionally
    unsexed_a unsexed_b}
one row for each stratum: stratum_name stratum_area n_tows
and then a table of length records, with a header row as follows:
stratum, tow, kg/km2, length, no_m, no_f, no_t
and a whole bunch of data rows.
```

The 'survey.sub' format is the same as this except that the final table of length records has header row

```
stratum, tow, subsample, kg/km2, subsample_kg/km2 length,
    no_m, no_f, no_t
```

and two additional columns ('subsample' = subcatch number, and 'subsample_kg/km2').

4.4 Precision of numbers in output files

The degree of precision (expressed as a number of significant figures or decimal places) for each of five types of output quantities is controlled by command `@output_precision` (Table 3). The default levels for each output type were set on the basis that c.v.s are unlikely to fall below 1% (and are usually much higher). Note that negative numbers can be used for decimal places (e.g., 12345.67 presented to -2 decimal places is 12300).

Table 3: Description of the quantities whose precision is controlled by command `@output_precision` and the default level of precision (a number of either significant figures or decimal places) assigned to each.

Quantity	Default precision	Description
density	4 sig. figs	all columns of type XYZ_kg, XYZ_kg/km, or XYZ_kg/km2 in station-data and stratum-data output files column kg/km2 in stratum_summary table column kg/km2 in catch-at-age output
biomass	4 sig. figs	all biomasses (including lower and upper bounds) in tables stratum_summary, biomass, sub_biomass_by_stratum, biomass_by_species, and biomass_by_species_stratum
LF_number	4 sig. figs	all scaled numbers in LF tables (i.e., all columns whose labels start with 'SN')
cv	1 dec. place	all columns in biomass or LF tables whose labels start with 'cv'
gain	0 dec. place	all numbers in gain tables (described in Section 4.1.1)

5. CALCULATIONS IN SURVCALC

This section contains detailed descriptions of the six types of calculation made by SurvCalc: of densities (Section 5.2), biomasses (Section 5.3), LFs (Section 5.4), phase-2 gain (Section 5.5), projected c.v.s (Section 5.6) and catch-at-age data (Section 5.7). These are preceded by definitions of the data and notation used (Section 5.1).

5.1 Data and notation

The data used in calculations come from three sources: the main input file (Section 3.1); flat files (Section 3.2); and the *trawl* database (see beginning of Section 3.1.1). The notation used in the calculations is explained in Tables 4 and 5.

Table 4: Variables used in calculations, their definitions, and how they are calculated or where they come from. Where a second variable is given in parentheses, the first variable is for the whole population and the second is the analogous variable for a sub-population. See Section 5.7 for additional notation used for task output_for_catch_at_age.

Variable	Definition	How calculated/where from
B (b)	overall biomass (t)	equation (7)
n	number of strata in survey	
m	number of strata with LF samples or zero catch	
CF	LF correction factor	equation (24)
<i>i</i> indexes the selected stations for the specified trip:		
t_i	time towed (h)	calculated from time_s, time_f, date_s, date_f
d_i	distance towed (n.mile)	Table 6
C_i (c_i)	catch weight (kg)	Table 6 (equations (10, 11))
w_i	width of tow (m)	Table 6
v_i	vulnerability	from @vulnerability
u_{vi}	vertical availability	from @vertical_availability
f_i	effective area fished (m^2)	$f_i = 1852d_iw_iv_iu_{vi}$ or @area_fished
AD_i (ad_i)	areal density ($kg.km^{-2}$)	equation (1) or station-data file
LD_i	linear density ($kg.km^{-1}$)	equation (2)
<i>s</i> indexes the selected strata for the specified trip:		
a_s	stratum area (km^2)	area_km2
n_s	number of selected stations in stratum s	
m_s	number of selected stations in stratum s where there is LF data or $C_i = 0$	
u_{as}	areal availability	from @areal_availability or stratum-data file
a'_s	population area (km^2)	$a'_s = a_s / u_{as}$
AD_s (ad_s)	mean areal density ($kg.km^{-2}$)	equation (3)
LD_s	mean linear density ($kg.km^{-1}$)	equation (4)
$AD_{s,LF}$	mean areal density ($kg.km^{-2}$) over stations with LF data or $C_i = 0$	equation (19)
p_s	proportion of catch taken in stations with LF data	equation (10)
B_s (b_s)	biomass (t)	equation (6)
CF_s	LF correction factor	equation (21)
K_s	LF scaling factor	equation (18)
<i>k</i> indexes sex (1 = male, 2 = female, 3 = unsexed, 4 = all):		
a_k, b_k, c_k	length-weight coefficients (calculates weight in g from length in cm)	from @lw_coeff

Table 4 continued:

j indexes the records in table `t_lgth` for station i and the specified trip and species (so the following variables are derived from attributes in this table)

Variable	Definition	How calculated/where from
p_{ij}	proportion sampled	percent_samp/100
N_{ijk}	number of fish by sex	no_m ($k=1$), no_f ($k=2$), no_a–no_m–no_f ($k=3$), no_a ($k=4$)
L_{ij}	length of fish (cm)	lgth
W_{ijk}	weight of fish (g)	$W_{ijk} = a_k L_{ij}^{b_k + c_k \ln L_{ij}}$
l indexes fish length		
W_{lk}	weight of fish of length l and sex k (g)	$W_{lk} = a_k l^{b_k + c_k \ln l}$
SN refers to scaled numbers of fish; MSN is the mean scaled numbers; SN' is scaled numbers corrected for consistency with biomass estimates and length-weight relationships		
SN_{ilk}	of length l and sex k at station i	equation (13)
SN_{slk} MSN_{slk} SN'_{slk}	of length l and sex k in stratum s	equations (14 or 16, 17, and 20)
SN_{lk} SN'_{lk}	of length l and sex k	equation (22, 23)
TN refers to total numbers (i.e., LF totals), which are sums over length of scaled numbers		
TN_{ik}	for station i	$TN_{ik} = \sum_l SN_{ilk}$
TN'_{sk}	for stratum s	$TN'_{sk} = \sum_l SN'_{slk}$ (& equation (28))
MTN_{sk}	for stratum s	$MTN_{sk} = \sum_l MSN_{slk} = \frac{\sum_{i, LF} TN_{ik}}{m_s}$
TN'_k	for the whole population	$TN'_k = \sum_l SN'_{lk}$

Table 5: Conventions used in summations. See Section 5.7 for a further convention used only for task output_for_catch_at_age.

Symbol	Range of summation
$\sum_i$	the n_s stations in stratum s
$\sum_{i, LF}$	the m_s stations in stratum s in which lengths were measured
$\sum_{i, LF2}$	stations in which <i>either</i> some lengths were measured <i>or</i> total catch was zero
$\sum_s$	the n strata in the survey
$\sum_{s, LF}$	the strata in which some lengths were measured
$\sum_{jk}$	all length records (j) and all sexes (k) for station i
$\sum_{jk, sub}$	values of j and k such that, for station i , the length L_{ij} and the sex k are within the required sub-population
$\sum_{j, l}$	values of j such that, for station i , the length $L_{ij} = l$
$\sum_l$	all lengths, l , measured at station i
$\sum_{l, sex}$	EITHER all lengths, l , for $k=4$, if a single set of length-weight coefficients is provided OR all lengths for $k = 1, 2, 3$, if length-weight coefficients are provided by sex

5.1.1 Note on subcatches

The notation (N_{ijk} , L_{ij}) and formulae (see below) used here for data from the table `t_lgth` are a bit more complicated than those of Vignaux (1994) because they allow for the possibility of what are called (in database *trawl*) subcatches, which is to say more than one length sample for a given species from a given station (e.g., one species in a catch may be divided roughly into two parts, ‘small fish’ and ‘others’, with each part being weighed separately, and having its own random length sample). This means that there may be two (or more) records in `t_lgth` with the same values of `trip_code`, `station_no`, `species`, and `lgth`. The sum $\sum_{j,l}$, used in calculating an LF for each tow (see below), combines such records.

In `trawlsurvey`, this summation is not done, and this produces anomalies in the output station LFs, as shown in the following example, in which there are two lines for fish of length 42 cm in station 79. However, this error affects only the station LFs; the stratum and overall LFs are OK.

```

stn  str  L  Nall  Nmale  Nfemale  Nunsexed  SNall  SNmale  SNfemale  SNunsexed
...
79 0017 41    12     8       4         0 338.11 225.41  112.70      0
79 0017 42     9     3       6         0 253.58  84.53  169.06      0
79 0017 42     1     1       0         0  14.77  14.77   0.00      0
79 0017 43    11     8       3         0 309.94 225.41   84.53      0
...

```

5.1.2 Note on stations and strata without LF data

There can be two types of stations (or strata) without LF data: A, those with no catch of the target species, and B, those with a non-zero catch from which no fish were measured. It is important to notice that both types are excluded from the summations $\sum_{i,LF}$ (for stations) and $\sum_{s,LF}$ (for strata), but that only type B is excluded in defining the variables m_s and m and in the summation $\sum_{i,LF2}$.

5.1.3 Excluding stations and strata

For tasks `calc_biomass`, `calc_LFs`, `calc_biomass_and_LFs`, and `phase_2_calc`, `SurvCalc` sometimes excludes some stations and/or strata because of inadequate data.

A station is excluded if A, it is not possible to calculate the fish density, AD_i , because of missing data (e.g., distance towed or width of tow missing, with no default provided) or B, the stratum it is in is excluded. Note that criterion A does not apply when the task is `calc_LFs` and `@LF_scaling` is not `numbers_per_km2` (because fish densities are not calculated in this case).

The criterion for excluding a stratum depends on the task. Stratum s is excluded if $n_s < 2$ for tasks `calc_biomass` and `phase_2_calc`, or if $m_s < 2$ for task `calc_biomass_and_LFs`.

When sub-population biomasses are calculated then any stratum in which $p_s < 0.3$ is excluded (but only for these biomasses) (see Section 5.3.1).

5.1.4 User preferences for fish-density variables

When command @preferences is used (see Section 3.1.4) it specifies the user's preferences for the variables d_i , w_i , and C_i . For each variable and station, the option used will depend on what data are available. The first option should be used unless the required data are not available, in which case the second option is used, unless the required data are not available, etc. All stations for which it is not possible to calculate all three variables (d_i , w_i , C_i) are dropped from the analysis (and an appropriate error message is included in the main output file). The options for each variable, and the associated values, are given in Table 6.

Table 6: Three variable with user-supplied calculation options, the associated sub-command of @preferences, and the value of the variable according to the option used.

Variable	Sub-Command	Option	Value
d_i	distance_towed	recorded_distance	distance in t_station
		from_lat_lon	calculated from following attributes in t_station: lat_s, NorS_s, long_s, EorW_s, lat_f, NorS_f, long_f, EorW_f
		from_recorded_speed from_constant_speed	speed in t_station x t_i @constant_speed x t_i
w_i	width_swept	recorded_doorspread	dist_doors in t_station
		constant_doorspread	@constant_doorspread
		recorded_wingspread	dist_wings in t_station
C_i	catch_weight	recorded	weight ¹ in t_catch
		calculated	$C_i = 0.001 \sum_{jk} N_{ijk} W_{ijk} / p_{ij}$

¹ Note that if there is no record in t_catch for the given species, station, and trip this means that $C_i = 0$.

5.1.5 Calculation of c.v.s

All c.v.s are calculated and presented as percentages. Unless otherwise stated, the c.v. of any variable, X , is calculated as $c.v.(X) = 100s.e.(X)/X$.

5.1.6 Use of length-weight coefficients

Length-weight coefficients, a_k , b_k , c_k , are not always required. They are used only in three situations:

- when C_i is calculated, rather than being a recorded value (Table 6),
- when sub-population biomasses are calculated (see equation (11), Section 5.3.1), and
- when LFs are calculated and scaled to numbers in population (see calculation of CF_s and CF in Section 5.4).

5.2 Calculating fish densities

One or both of two types of densities may be calculated: areal (kg/km²) or linear (kg/km). The former are always used for calculating either biomass or gains. The latter (not defined if @survey_type is pot) are calculated only if requested (with commands @station_data_columns and/or @stratum_data_columns) as output columns in station or stratum tables.

When a station-catch input file is used (see Section 3.2) the station densities are read directly from this file: AD_i from columns with names like XYZ_kg_km2; LD_i from columns with names like XYZ_kg_km. Otherwise, they are calculated as

$$AD_i = 10^6 C_i / f_i \quad (1)$$

$$LD_i = 1000 C_i / (1.852 d_i u_i v_i) \quad (2)$$

Stratum mean densities are calculated as

$$AD_s = (\sum_i AD_i) / n_s \quad (3)$$

$$LD_s = (\sum_i LD_i) / n_s \quad (4)$$

The s.e.s of areal densities are given by

$$\text{s.e.}(AD_s) = \left(\left[\sum_i (AD_i - AD_s)^2 \right] / [n_s (n_s - 1)] \right)^{0.5} \quad (5)$$

5.3 Calculating biomasses

Stratum and overall biomasses are calculated as

$$B_s = AD_s a'_s / 1000 \quad (6)$$

$$B = \sum_s B_s \quad (7)$$

and their standard errors are calculated as

$$\text{s.e.}(B_s) = \text{s.e.}(AD_s) a'_s / 1000 \quad (8)$$

$$\text{s.e.}(B) = \left[\sum_s \text{s.e.}(B_s)^2 \right]^{0.5} \quad (9)$$

5.3.1 Calculating sub-population biomasses

The first step in calculating sub-population biomasses for each stratum is the calculation of sub-population catches, c_i , for each station.

In the special case of a stratum with zero catch (i.e., $\sum_i C_i = 0$), the sub-population catch is set to 0 for all stations in that stratum.

For all other strata, a check is done to see whether there are enough LF data to warrant the calculation of sub-population catches. This involves the calculation of the quantity

$$p_s = \sum_{i,LF} C_i / \sum_i C_i \quad (10)$$

which is the proportion of the catch in stratum s that is from stations with LF data. Sub-population biomass estimates will be unreliable in strata where p_s is low. Any strata with $p_s < 0.3$ are dropped from the calculation of sub-population biomasses and a warning message is included in the output file. Strata with $0.3 \leq p_s < 0.5$ are included in the calculation of sub-population biomasses but a warning message is printed.

Where $p_s \geq 0.3$, the sub-population catch for station i is estimated as

$$c_i = \left[C_i \sum_{jk,sub} N_{ijk} W_{ijk} / p_{ij} \right] / \left[\sum_{jk} N_{ijk} W_{ijk} / p_{ij} \right] \quad (11)$$

for each station with LF data, and

$$c_i = C_i \left[\sum_{i',LF} ad_{i'} \right] / \left[\sum_{i',LF} AD_{i'} \right] \quad (12)$$

for stations without such data.

The remaining calculations (of densities, ad_i and ad_s , biomasses, b_s and b , and s.e.s, for ad_s , b_s and b) use the same equations as for population biomass (i.e., equations 1, 3, and 5-9) except that each population variable is replaced by its sub-population analogue (as defined in Table 4).

5.4 Calculating LFs

The equation for calculating SN_{ilk} , the scaled LF for the i th tow, depends on which option has been selected with command `@LF_scaling`:

$$SN_{ilk} = \begin{cases} \sum_{j,l} N_{ijk} & \text{unscaled} \\ \frac{1000 \sum_{j,l} (N_{ijk} / p_{ij})}{\sum_{i'} \left[\sum_{j,l} (N_{ijk} / p_{ij}) \right]} & \text{numbers_per_1000} \\ \sum_{j,l} (N_{ijk} / p_{ij}) & \text{numbers_per_tow} \\ \sum_{j,l} (N_{ijk} / (p_{ij} t_i)) & \text{numbers_per_hour} \\ \sum_{j,l} (10^6 N_{ijk} / (p_{ij} f_i)) & \text{numbers_per_km2 or} \\ & \text{numbers_in_population} \end{cases} \quad (13)$$

For stations with zero catches, $SN_{ilk} = 0$ for all l and k .

How the stratum and overall LFs are calculated depends on the task. If it is `calc_LFs` these are simple (i.e., unweighted) sums or averages, with no correction for stations or strata without LFs:

$$SN_{slk} = \begin{cases} \sum_{i,LF} SN_{ilk} & \text{for unscaled LFs} \\ \left(\sum_{i,LF} SN_{ilk} \right) / m_s & \text{otherwise} \end{cases} \quad (14)$$

$$SN_{lk} = \begin{cases} \sum_{s,LF} SN_{slk} & \text{for unscaled LFs} \\ \left(\sum_{s,LF} SN_{slk} \right) / m & \text{otherwise} \end{cases} \quad (15)$$

These equations differ from those in trawlsurvey, where SN_{slk} and SN_{lk} are always calculated as sums for a `calc_LFs` analysis. That is, $SN_{slk} = \sum_{i,LF} SN_{ilk}$ and $SN_{lk} = \sum_{s,LF} SN_{slk}$.

When the task is `calc_biomass_and_LFs` the stratum LFs are calculated by averaging the station LFs, scaling using the population area, and then adjusting for strata with no LF data

$$SN_{slk} = MSN_{slk} K_s \quad (16)$$

where
$$MSN_{slk} = \left(\sum_{i,LF} SN_{ilk} \right) / m_s \quad (17)$$

$$K_s = \frac{a'_s AD_s}{AD_{s,LF}} \quad (18)$$

and
$$AD_{s,LF} = \left(\sum_{i,LF} AD_i \right) / m_s \quad (19)$$

They are then corrected so that the estimated numbers of fish are consistent with the estimated biomass and length-weight relationships

$$SN'_{slk} = SN_{slk} CF_s \quad (20)$$

where
$$CF_s = 10^6 B_s / \sum_{l,sex} SN_{slk} W_{lk} \quad (21)$$

Overall LFs are calculated using

$$SN_{lk} = \sum_{s,LF} SN_{slk} \frac{B}{\sum_{s,LF} B_s} \quad (22)$$

(note that for surveys in which there are LF data in all strata this equation becomes a simple sum, $SN_{lk} = \sum_s SN_{slk}$) and then corrected for consistency with the estimated biomass and length-weight relationships

$$SN'_{lk} = SN_{lk} CF \quad (23)$$

where
$$CF = 10^6 B / \sum_{l,sex} SN_{lk} W_{lk} \quad (24)$$

5.4.1 Calculating c.v.s for LFs

C.v.s for LFs are calculated only when the task is `calc_biomass_and_LFs`, and then only for corrected LF totals, TN'_{sk} (for each stratum) and TN'_k (for the whole population), using the equations

$$c.v.(TN'_{sk}) = 100 s.e.(MTN_{sk}) / MTN_{sk} \quad (25)$$

and

$$\text{c.v.}(TN'_k) = \frac{100 \left(\sum_{s,LF} \text{s.e.}(MTN_{sk})^2 K_s^2 \right)^{0.5}}{\sum_{s,LF} MTN_{sk} K_s} \quad (26)$$

where

$$\text{s.e.}(MTN_{sk}) = \left(\frac{\sum_{i,LF2} (TN_{ik} - MTN_{sk})^2}{m_s (m_s - 1)} \right)^{0.5} \quad (27)$$

The derivation of equations (26-27) is worth explaining. These c.v.s take into account only the uncertainty arising from the variation in numbers between stations in the same stratum. All other sources of uncertainty are ignored. Thus, because we can write

$$TN'_{sk} = MTN_{sk} K_s CF_s \quad (28)$$

and we ignore uncertainty in K_s and CF_s , $\text{c.v.}(TN'_{sk}) = \text{c.v.}(MTN_{sk})$, which leads to equation (25). Similarly, we can write

$$TN'_k = \left(\sum_{s,LF} MTN_{sk} K_s \right) \left[\frac{B}{\sum_{s,LF} B_s} CF \right] \quad (29)$$

but we ignore uncertainty in both the K_s and the term in square brackets. Thus $\text{c.v.}(TN'_k) = \text{c.v.}\left(\sum_{s,LF} MTN_{sk} K_s\right)$, which leads to equation (26).

5.5 Calculating phase-2 gains

Gains are calculated only when the task is `phase_2_calc`, and the calculations are based on the equations in Francis (1984). When these calculations are done, it is assumed that the selected stations (i.e., those defined by commands in Section 3.1.1, or included in the station flat file, if there is one) are all the phase-1 stations in a 2-phase survey. Sometimes, for logistical reasons, the calculations will be done before all phase-1 stations are completed and/or after some phase-2 stations have been done. In this situation, the calculations should use all completed phase-1 stations but *not* any of the phase-2 stations (to use phase-2 stations in the calculations would effectively change the survey from 2-phase to multi-phase, and the statistical behaviour of multi-phase surveys of this type are not known).

A table of gains will be calculated for each species in `phase_2.species`. If sub-populations are defined for any of these species, then an additional table of gains will be calculated for each sub-population.

Each table of gains will have n rows (i.e., one per stratum) and `phase_2.n_station` columns. Let G_{se} be the relative gain (i.e., the estimated reduction in the biomass variance) associated with adding the e th extra station to the s th stratum. To calculate G_{se} , we need first to calculate AD_s (and ad_s if sub-populations are involved) and its s.e. (using equations 1, 3, and 5). The formula for G_{se} depends on whether `phase_2.algorithm` is `mean_squared` or `variance`:

$$G_{se} = \begin{cases} \frac{(a'_s AD_s)^2}{(n_s + e - 1)(n_s + e)} & \text{mean_squared} \\ \frac{(a'_s \text{s.e.}(AD_s))^2 n_s}{(n_s + e - 1)(n_s + e)} & \text{variance} \end{cases} \quad (30)$$

(for sub-populations, replace AD_s by ad_s in this equation).

Before being output, each table of gains is to have maximum value 100 using the equation

$$G_{se} = 100G_{se} / \max_s (G_{s1}) \quad (31)$$

5.6 Calculating projected c.v.s

Projected c.v.s are calculated for all biomasses (including those for sub-populations) when the task is `calc_biomass` or `calc_biomass_and_LFs` and there is a `@projected_cvs` command block (see Section 3.1.4) in the main input file. In these calculations, n_s is the number of stations already completed in stratum s , and e_s is the number of stations yet to be completed (the latter being specified in the `@projected_cvs` command block). For the projections it is assumed that, in each stratum, the mean and variance of the catch rates for the remaining stations will be exactly the same as for the existing stations.

The calculations use the same equations as are used in the calculation of the usual biomass c.v.s (i.e., equations (8) and (9)) except that, for strata that are not yet completed (i.e., those specified in the `@projected_cvs` command block), $\text{s.e.}(AD_s)$ (or $\text{s.e.}(ad_s)$ in the case of sub-populations) in equation (8) is replaced by $\text{s.e.}_{\text{proj}}(AD_s)$ (or $\text{s.e.}_{\text{proj}}(ad_s)$) where these are defined by

$$\text{s.e.}_{\text{proj}}(AD_s) = \text{s.e.}(AD_s) (n_s / (n_s + e_s))^{0.5} \quad (32)$$

and
$$\text{s.e.}_{\text{proj}}(ad_s) = \text{s.e.}(ad_s) (n_s / (n_s + e_s))^{0.5} \quad (33)$$

5.7 Output for catch-at-age

With task `output_for_catch_at_age` SurvCalc must read additional material from the database (beyond what is given in Table 1) and may have to do some additional calculations.

The first additional material that must be read from the database is column `subcatch_no`, from database table `lgth`. If this column is always equal to 1 there are no subcatches, so the output will be in format ‘survey’ and no additional calculations are required. The column `kg/km2` in the output file (see Section 4.3) will contain AD_i .

If there are any subcatches, the output will be in format ‘survey.sub’, column `subsample` in the output file will be `subcatch_no` from table `lgth`, and column `subsample_kg/km2` will contain the subcatch catch rate AD_{ic} , where c indexes the subcatches (for the species in question) at station i , and $AD_{ic} = 10^6 C_{ic} / f_i$.

Where C_{ic} comes from depends on what option is being used for `catch_weight` (as specified in the `@preferences` command – see Section 3.1.4). With option `recorded`, C_{ic} is taken directly from column `weight` in table `t_subcatch` (so SurvCalc must read columns `trip_code`, `station_no`, `species`, `subcatch_no`, and `weight` from this table). With option `calculated`, $C_{ic} = 0.001 \sum_{jk,c} N_{ijk} W_{ijk} / p_{ij}$, where $\sum_{jk,c}$ means the sum over all sexes (k) and all length records (j) in station i for which `subcatch_no` = c .

6. SurvCalc AND 2-PHASE SURVEYS

SurvCalc, and the associated R library, can be used in three ways to help in 2-phase surveys.

First, the R function `allocate` can be used in designing the survey (see Appendix 2). One aspect of the design is the percentage of stations to be done in phase 1 of the survey. Note that although Francis (1984) recommended this be 75%, the current recommendation, based on much more extensive simulations, is 90% (Francis 2006).

Second, task `phase2_calc` of SurvCalc can be used at the end of phase 1 to help in determining the allocation of phase-2 stations to strata. For example, suppose our target species is hoki and, at the end of phase 1, we decide there is sufficient remaining survey time to complete six phase-2 stations. Then we could insert the command block

```
@phase_2 HOK
n_stations 6
```

into our main input file, and run `SurvCalc -2 > myfile`.

Within the output file, `myfile`, there will be a table of gains for hoki which might look like

stratum	1	2	3	4	5	6
0001	19	15	12	10	8	7
0002	<u>62</u>	44	33	26	21	17
0003	<u>100</u>	<u>60</u>	40	29	21	17
0004	<u>96</u>	<u>64</u>	<u>46</u>	34	27	21
0005	43	32	25	20	16	14

In this example, the six biggest numbers in the table have been underlined to help illustrate how the table of gains should be interpreted. This shows us that the optimum allocation for 6 phase-2 stations is one station in stratum 0002, two stations in stratum 0003, and 3 stations in stratum 0004. If it turned out that there was time only for five phase-2 stations, then we should ignore the smallest underlined number, and thus do only two extra stations in stratum 0003.

The third way that SurvCalc can be useful in 2-phase surveys is in evaluating alternative phase-2 allocations using the command `@projected_cvs`. If, for example, hake was also a target species in the above survey, we would need to add the command block

```
@phase_2 HAK
n_stations 6
```

to our main input file before running `SurvCalc -2 > myfile`. Then `myfile` would contain two tables of gains (one for hoki, and one for hake), from which we could construct a table like the following showing two alternative phase-2 allocations

stratum	phase-2 allocation	
	for HOK	for HAK
0001	0	2
0002	1	2
0003	2	1
0004	3	1
0005	0	0

So, which phase-2 allocation should we use: the one for hoki, the one for hake, or some compromise between the two? To answer this question, the first thing to do is to find out

what c.v.s we might expect to get for the hoki and hake biomass estimates using each of the above allocations. To evaluate the hoki allocation we would add the command block

```
@projected_cvs
strata          0002 0003 0004
extra_stations   1    2    3
```

and run `SurvCalc -b > myfile`. As well as calculating all the phase-1 biomass estimates and c.v.s, this run will produce a table like

```
species population  projected_cv
HOK                all           18.3
HAK                all           25.7
...
```

giving estimates of the likely c.v.s at the end of the survey if we were to use the hoki allocation. We can repeat this run, changing the allocation in the `@projected_cvs` command block until, by trial and error, we find the most suitable phase-2 allocation.

Another reason for considering alternative phase-2 allocations is that some strata may be more expensive (in terms of steaming time) to return to than others. Suppose, for example, that in the above example stratum 0002 was so far away that including it in phase 2 would mean that there was time for only five phase-2 stations. Then we could use `@projected_cvs` to compare the projected c.v.s from the following two alternative allocations.

stratum	phase-2 allocation	
	with str 0002	without str 0002
0001	0	0
0002	1	0
0003	2	2
0004	2	3
0005	0	1
Total	5	6

The first of these allocations is what we get from the above table of gains if we allocate only five phase-2 stations, and the second is the result of allocating six phase-2 stations but ignoring stratum 0002.

Some practical considerations

In theory, the rule is that our phase-2 calculations (using either task `phase2_calc`, or task `calc_biomass` with command `@projected_cvs`) should be done only after we have finished all phase-1 stations, and before we have started phase 2. However, it is perfectly acceptable to break this rule in some situations.

Remote strata provide a good reason to do phase-2 calculations before the end of phase 1. After completing the phase-1 stations in a remote stratum we might want to use task `phase2_calc` to work out, on the basis of the phase-1 stations completed to date, whether we are likely to need any phase-2 stations in that stratum and, if so, approximately how many. If it looks like some phase-2 stations will be needed in this stratum it is quite permissible to do them immediately, before leaving the area.

Another reason to break the rule is if, part-way through phase 2, it becomes clear that the actual number of phase-2 stations will differ from what was assumed, so that there is a need to modify the original phase-2 allocation.

There is one rule about phase-2 calculations that should never be broken. That is, you must never include catches from phase-2 stations when doing phase-2 calculations. Suppose, for example, that during phase 1 you added two phase-2 stations (numbers, 32 and 33, say) in a remote stratum. Then these stations must be excluded from any subsequent phase-2 calculations. This could be done using subcommand `t_station station_no not in (32,33)` of command `@where`. If you were already using this subcommand (say to exclude stations with poor gear performance) then this restriction can be added as `t_station gear_perf < 3 and station_no not in (32,33)`. Similarly, suppose you had completed 100 phase-1 stations (stations 1–100) and were part-way through phase 2 when you wanted to repeat the phase-2 calculations. Then you could use subcommand `t_station station_no < 101` to exclude phase-2 stations from these calculations.

7. REFERENCES

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8. APPENDIX 1: COMMAND BLOCK FORMAT

The main input file for SurvCalc uses a *command-block format* similar to that used in CASAL (Bull et al. 2008). That is, this file consists of a series of command-blocks, which may occur in any order. Each command-block either consists of a single command (starting with the symbol @) and its arguments, or a command (starting with @) and an optional label and one or more subcommands, i.e.,

@command *arguments*
or

@command [*label*]
[subcommand *arguments*]
[subcommand *arguments*]
[...]

Blank lines are ignored, as is extra white space between arguments. Comments beginning with '#' are ignored. If you want to remove a group of commands or subcommands using '#', then comment out the whole block, not just the first line. Alternatively, you can comment out an entire block by placing curly brackets around the text that you want to comment out. Put in a '{' as the first character on the line to start the comment block, then end it with '}'. All lines (including line breaks) between '{' and '}' inclusive are ignored. (These should ideally be the first character on a line, but if not, then the entire line will be treated as part of the comment block.)

Don't put extra white space before a @ character (which must also be the first character on the line). Make sure the file ends with a carriage return. Commands and subcommands must consist of letters and/or underscores, and must not contain a full-point ('.').

There is no need to mark the end of a command block. This is automatically recognised by either the end of the file or the start of the next command block, which is marked by the @ on the first character of a line.

Also note that the commands, sub-commands, and arguments in the parameter files are case sensitive.

Some commands can never have subcommands (such as @survey_type). If a command has no subcommands, then it has to have arguments, which are placed on the same line as the command. All other commands have no arguments, but have subcommands instead. Some commands can be used multiple times and if they are they must have a different label each time (see Table 2).

9. APPENDIX 2: THE SURVCALC R LIBRARY

This library can be downloaded from <https://one.niwa.co.nz/display/FISHMOD/SurvCalc>. It contains two main functions: `input.SurvCalc` and `allocate`.

input.SurvCalc

Function `input.SurvCalc` simply reads the main output file of SurvCalc into R. Thus if you have a SurvCalc output file `myout` in directory `H:\proj\hok` you can read this into the R object `hokout` using the R command

```
hokout <- input.SurvCalc('H:/proj/hok/myout')
```

or, alternatively

```
hokout <- input.SurvCalc('myout', 'H:/proj/hok')
```

The object `hokout` is a list of lists that is structured hierarchically, by trips and then species. For example, all the output concerning trip `buc8501` in `myout` is contained in the list `hokout$buc8501`. To see what tables are there, simply type `names(hokout$buc8501)`, which may return something like

```
[1] "SELECTIONS" "RUN PARAMETERS" "ORH" "OEO"
```

`hokout$buc8501$SELECTIONS` would contain information about the strata and stations selected for this trip, and `hokout$buc8501$"RUN PARAMETERS"` would contain the user-supplied preferences for distance towed, width swept, etc.

Similarly, typing `names(hokout$buc8501$ORH)` lists the output tables for ORH in trip `buc8501`. For a description of all tables that can occur in a main output file from SurvCalc see Section 4.1.

allocate

Function `allocate` is useful at the survey design phase. It uses historical data from previous surveys in the same area to work out how many stations should be allocated to each survey stratum. It can be used in two different ways, as illustrated by the following calls.

```
allocate(constraint='cv', limit=20, strata=mystrat, data=mydat)
allocate(constraint='stations', limit=20, strata=mystrat, data=mydat)
```

The first call calculates the minimum number of stations, and their allocation to strata, to achieve a c.v. of 20%, whereas the second calculates the best allocation given that there will be a total of 20 stations.

In both cases, `mystrat` should be a dataframe containing stratum information in the format

stratum	area	minimum	maximum
0001	1245	3	20
0003	347	3	15
...			

with the last two columns specify the minimum and maximum number of stations allowed in each stratum, and `mydat` should be a dataframe containing the historical data in the format

```
stratum survey    catch
0001    tan0303     3.7
0001    tan0303     2.5
0001    tan0303    12.1
0002    tan0303     4.2
0002    tan0303     2.8
...
0001    tan0402     5.3
...
```

where the last column contains catch rates (e.g., t km^{-1} or t km^{-2} , etc).

The output from `allocate` is a list with three components: `cv` (the estimated c.v. achieved by the calculated allocation); `stations` (the total number of stations); and `allocation` (the allocation of those stations to the strata). For example, the output from the first call above might look like the following

```
$cv
[1] 19.8

$stations
[1] 32

$allocation
  stratum stations
1    0001        17
2    0002        12
3    0003         3
```

which would indicate that 32 stations (with 17, 12, and 3 in strata 0001, 0002, and 0003, respectively) could be expected to achieve a c.v. of 19.8%.

It is not uncommon for surveys to have separate target c.v.s for two or more species. In this case we would need to construct a different dataframe of historical catch rates for each species, and run `allocate` with each of these in turn. In this case it is of interest to find out what c.v. would be expected for one species if we used the optimal allocation for the other species. This is easily done by modifying the strata dataframe so that for each stratum the minimum and maximum numbers of stations are both set equal to the optimal number.

It's important to understand that `allocate` is a very simple function, which makes no allowance for the very real possibility that the distribution of fish in the survey being planned may be quite different from that in the historical surveys. For this reason it is prudent to be conservative in its use. For example, it is common to use `allocate` to determine a station allocation that will achieve the target c.v.s in phase 1 of a 2-phase survey. Then, if fish distributions during the survey differ substantially from those in previous surveys, so that the target c.v.s are not met during phase 1, there is still a possibility that they will be met by the end of phase 2.